

Design and Optimization of a Voice Coil Actuator for Precision Motion Applications

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There has been increasing attention to the use of voice coil actuators (VCAs) for precision motion applications. In this paper, a detailed design of a cylindrical VCA is presented. Different options for the overall configuration are evaluated according to various criteria and design variables are defined for the chosen configuration. After that, optimization parameters are derived to maximize the performance in a precision motion application by maximizing the acceleration and minimizing the heat dissipation. Design parameters are optimized using finite element analysis to evaluate the magnetic properties. Optimization is carried out using the bulk volume of coil, which allowed electrical properties to be later characterized via the selection of wire gauge. Calculations for evaluating resistance, inductance, current drain, and voltage supply requirements for dc and dynamic cases are presented. Magnetic field predictions and formulations used in force calculations are verified in experiments.

Index Terms—Coil wire gauge, finite element analysis, optimization, precision motion systems, voice coil actuator.

I. INTRODUCTION

A. Background

VOICE COIL actuators (VCAs) are a class of electromagnetic devices that work on the principle of Lorentz forces to generate actuation within relatively short stroke lengths proportional to their own length. Their name derives from their common use in loudspeakers. VCAs are particularly useful in precision motion applications, for they are free from mechanical hysteresis, force or torque ripple, and backlash, due to their non-contact and continuous operation [1], [2]. Their simplistic design and cog-free motion makes them feasible, although their stroke length is typically shorter than multi-phase electromagnetic linear motors. Owing to the large versatility in the design of VCAs regarding size, shape, stroke length, configuration, and capacity, several different uses of them in precision motion systems have emerged [3]–[13]. Depending on the design requirements, VCAs can be made to directly actuate a stage resting on some type of support structure, such as an air bearing [3]–[5], [7] or a platform with a ground V-groove [6]. Also, they can be used to provide the lifting force for a system made to be levitated, while also providing the differential forces for maneuvering [12]. For reasons of preloading the system and providing a motion guide/bearing, VCAs can be designed to operate indirectly by means of a flexure mechanism [8]–[11]. It is possible to couple VCAs in parallel with other types of actuators, to enhance the positioning performance [3], [6], [9], [13]. In [3], the VCA-driven stage is mounted on a ball-screw-driven stage to provide the fine motion. In contrast, VCAs provide the coarse motion for piezoelectric (PZT) actuators in [6] and [9], while PZT motion is further refined by a thermal actuator in [13]. Although it is also largely dependent on the skill of the designer and the performance of other components,

motion systems employing only VCAs for actuation have demonstrated competence in positioning accuracy. The tracking system for magnetic recording has 0.5 nm positioning accuracy (after the 8-nm sensor noise is filtered out) [4], the planar motion table with ± 10 -mm range has 1 nm positioning resolution on each axis [7], and the single axis nanopositioning system could achieve tracking with 10 nm rms error [11].

Optimization of VCAs has been investigated in several studies [14]–[18], with the overall shape and structure of the VCA varying according to the applications as high-precision motion stage [15], [16], optical disk drive [18], or an exemplary design case [14], [17]. In [15], the only requirement from the VCA is the ability to actuate the system with regard to the forces demanded by the flexure bearing. The requirement to limit heat dissipation is handled indirectly through constraints. In [16], the cost function is a combination of dynamic response, acceleration, and heat dissipation, while in [18], it is aimed to minimize vibratory disturbances on the system. For the evaluation of magnetic fields, finite element (FE) analysis [17], [18], magnetic equivalent circuit (MEC) method [15], [16], or a combination of the two [14] can be used. Method of optimization also differs among different works such as manifold-mapping [14], sequential quadratic programming (SQP) [15], genetic algorithm (GA) [16], level set (LS) method [17], and design of experiments (DOE) [18]. The analysis method has been verified in experiments in [15], [16], and [18].

B. Contribution

In this paper, a VCA for precision motion applications is designed and optimized using novel optimization objectives. The objectives do not directly follow from past studies, but reflect similar needs of precision motion applications. The primary difference of them is that they are derived from wire-gauge-independent formulations for electromagnetic quantities. Coil wire gauge selection, decoupled this way from performance optimization, is investigated for its effects on the electrical characteristics of the VCA.

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Precision motion systems require high bandwidths, which require high acceleration capacity, for increased controller accuracy. When the *moving magnet* mode is used, as assumed in this paper, the mass of the steel core adds to the total moving mass. In *acceleration per current density* in Section III-A1, acceleration is maximized for this total mass. When operated in *moving magnet* mode, the VCA resembles a moving magnet actuator (MMA), with the difference that MMA has reduced moving mass due to the separation of iron and the magnet [19]. The *moving coil* mode is not chosen as it introduces friction and stiffness from lead wires, which is undesirable in precision motion applications. However, moving coil has reduced moving mass, as the coil is generally lighter than the steel core. Another requirement for precision motion applications is minimum thermal distortion, since precision systems may fail in the presence of them. Square of acceleration per heat dissipation is maximized in [16] and [18]. Motor constant in Section III-A2 maximizes force produced per unit square root of heat dissipation. In the aforementioned VCA optimization studies [14]–[18], wire gauge for the coil either is a preset or appears as a variable inside the performance criteria. In this paper, optimization objectives are derived independently of coil wire size. Therefore, coil design can be made separately, according to desired electrical characteristics.

C. Organization

This paper is organized as follows. The VCA design problem is detailed in Section II. Section III presents the optimization procedure. In Section IV, effects of coil design on the electrical characteristics of the final system are presented. In Section V, experimental verification of the finite element analysis (FEA) technique and force calculations are presented. Concluding remarks are presented in Section VI.

II. DESCRIPTION OF THE VCA

A. Configuration Options for the Core

A cylindrical VCA comprises an iron core and an engaging coil as shown in Fig. 1. The equation of Lorentz force resulting from a current passing through a conductor of finite length is given as [20]

$$\vec{F} = \int_U I d\vec{u} \times \vec{B} \quad (1)$$

where $\vec{F}$ represents the Lorentz force, I represents the uniform current, U represents the conductor length, $\vec{B}$ represents the magnetic flux density vector, and $d\vec{u}$ is the variable of integration for conductor length.

In order to produce a Lorentz force in the direction of the axis of symmetry, a component of the magnetic flux in the gap perpendicular to the coil axis, as indicated by the large arrows in Fig. 1, is needed. The overall path of the streamlines that would produce such a gap flux field is also presented. In this case, the permanent magnet can be placed in any of the locations marked as A, B, or C. Each location corresponds to a different direction of magnetization and shape for the magnet. Location A requires a radially magnetized annular magnet. Such magnets are costly and difficult to obtain.

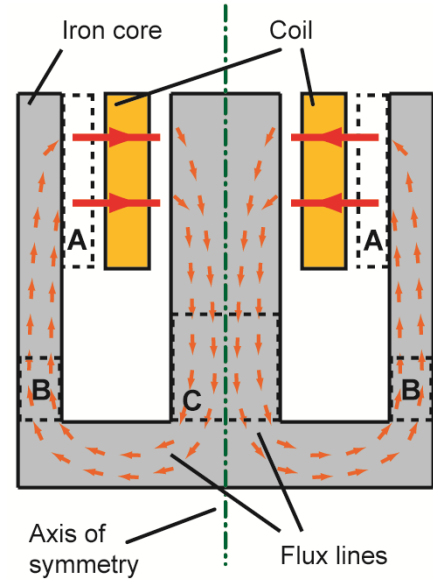


Fig. 1. Overview of the cylindrical VCA. A, B, and C are possible locations for the permanent magnet.

Approximating it with a circular array of separate magnets makes the assembly relatively more complicated. Location B requires an axially magnetized ring magnet, whereas Location C requires an axially magnetized cylindrical magnet. Being easy to obtain in both standard and custom sizes, magnets B and C have similar availability. The ring magnet of Location B can be easily centered using the already available outer cover for the iron core manufactured to a tight tolerance. On the other hand, in order to align the cylindrical magnet of Location C, a separate annular piece has to be produced with accurate dimensions in both inner and outer sides. For the sake of the practical assembly, Location B is chosen in this paper. Neodymium-iron-boron (NdFeB) magnets have higher remanence, coercivity, and energy product compared with Alnico, ceramic (ferrite), and rare earth cobalt magnets, which is the reason they are preferred in this paper. Furthermore, as the demagnetization curve of NdFeB magnets in the second quadrant of the B – H plane is mostly linear and their recoil permeability closely follows its slope, these magnets do not lose permanent magnetization during assembly or due to coil magnetic fields. If an alnico or a ceramic (ferrite) magnet was used, the additional benefit of Locations B and C would be to place the coil relatively far from the magnet, and thus prevent demagnetization due to coil magnetic fields [21], [22]. A shorted turn in the gap could have been considered to decrease inductance, regardless of the resulting reduction in gap length, but it would also result in additional heating. The presented possible configurations all suffer from reduction in thrust force, as the coil disengages from the core, which can be counteracted by using actuators in couples at opposing ends of each motion axis.

B. Design Variables

Outline of the VCA design subject to optimization is shown in Fig. 2. The summary of design variables is presented

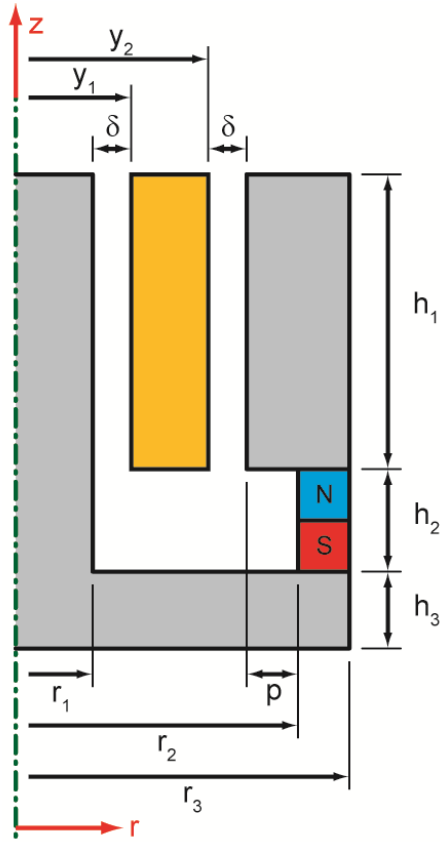


Fig. 2. VCA design variables.

TABLE I
SUMMARY OF VCA DESIGN VARIABLES

Symbol	Description	Value (mm)
h_1	stroke length	20
h_2	magnet height	8
h_3	core base height	-
r_1	core inner radius	-
r_2	magnet inner radius	17.5
r_3	magnet outer radius	22.5
p	reduction in gap	-
δ	clearance	-
y_1	coil inner radius	$r_1 + \delta$
y_2	coil outer radius	$r_2 - p - \delta$

in Table I. Among these variables, some are preset to values arising from design limitations. Stroke length h_1 is determined from the desired range of motion. Scale of the VCA r_3 is determined with respect to the force requirements of the application. Its exact value is refined according to magnet availability. Dimensions r_2 and h_2 follow from the chosen magnet. The remaining set of variables, $\{r_1, p, \delta, h_3\}$, are determined with respect to optimization objectives in Section III. In order to reduce leakage fluxes, a common practice especially in loudspeakers [23] is to make the diameter of pole pieces slightly smaller than the magnet. Also, a chamfer on the

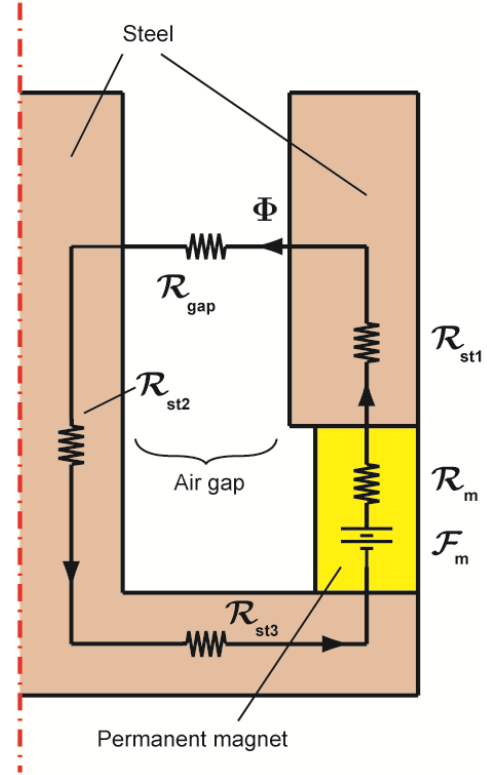


Fig. 3. MEC model of the VCA.

outer radius of the base plate can be used to reduce end leakage. These features were not utilized, as they complicate the in-house assembly of the VCA. They are addressed in the Appendix.

C. Magnetic Properties

MEC is a method that allows analytical expressions for the magnetic flux densities to be derived for any point in a magnetic system, using approximations with lumped circuit elements. Where there are considerable leakage fluxes, these can be accounted for by estimating the permeance of probable flux paths using simple shaped volumes [24], [25]. With experience and by employing sufficient level of detail in approximations, satisfactory results can be obtained if the structure is not overly complicated. In this paper, MEC analysis is carried out to demonstrate the relationship between magnetic field distributions and key design variables in the VCA, while the main results are obtained by FEA. Therefore, leakage fluxes are not considered in the MEC model, as shown in Fig. 3. The expressions for magnetic circuit elements on the flux path are calculated [26] and summarized in Table II.

Sample values for magnetic circuit elements with $\{\delta = 0.5, h_3 = 6, r_1 = 10, p = 2\}$ are also given. The expression for the magnetic flux is given as

$$\Phi = \frac{F_m}{R_m + R_{\text{gap}} + R_{\text{st1}} + R_{\text{st2}} + R_{\text{st3}}} \quad (2)$$

where F_m is the magnetomotive force (MMF) of the magnet, R_m is the internal reluctance of the magnet, R_{gap} is the

TABLE II
EXPRESSIONS FOR MAGNETIC CIRCUIT ELEMENTS AND SAMPLE VALUES

Variable	Expression	Value
B_r	remance of permanent magnet (p.m.)	1.28 (T)
μ_o	permeability of air	$4\pi \times 10^{-7}$ (H/m)
μ_m	permeability of p.m.	$1.1 \times \mu_o$
μ_{st}	permeability of steel	$4000 \times \mu_o$
F_m	$\frac{B_r h_2}{\mu_m}$	7408 (A.t)
R_m	$\frac{h_2}{\mu_m \left[\frac{\pi}{2} (r_3^2 - r_2^2) \right]}$	1.842×10^7 (1/H)
R_{gap}	$\frac{1}{\mu_o \pi h_1} \ln \left(\frac{r_2 - p}{r_1} \right)$	5.551×10^6 (1/H)
R_{st1}	$\frac{h_1/2}{\mu_{st} \left[\frac{\pi}{2} (r_3^2 - (r_2 - p)^2) \right]}$	4761 (1/H)
R_{st2}	$\frac{h_1/2 + h_2 + h_3/2}{\mu_{st} \left(\frac{\pi}{2} r_1^2 \right)}$	26597 (1/H)
R_{st3}	$\frac{1}{\mu_{st} \pi h_3} \ln \left(\frac{r_3}{r_1} \right)$	8559 (1/H)

reluctance of the air gap and $R_{st1...st3}$ are the reluctances of steel parts. Using this expression, gap flux density at the middle of the gap is evaluated as $B = \Phi/A_{gap} = 0.385$ T. The value found from this MEC analysis should be taken only as a guideline since it does not take the following into account:

- 1) fringing, which diverts the flux from its desired radial path;
- 2) losses from leakage fluxes.

It is seen in Table II that the largest reluctance originates from the air gap. The reluctance of the magnet is an imaginary one which approximates the $B-H$ curve of NdFeB. In order to increase the flux density in the gap, gap length has to be minimized. On the other hand, gap length also determines the volume of coil, which can be accommodated inside the iron core, hence is directly proportional to the achievable force.

III. OPTIMIZATION

The optimization problem can be stated as follows. Design variables $\{r_1, h_3, p, \delta\}$ of the VCA have to be optimized, such that acceleration per current density, σ_{acc} is maximized, and motor constant, K_M is maximized. The optimization objectives σ_{acc} and K_M are derived in the following.

A. Optimization Objectives

1) *Acceleration per Current Density*: Equation (1) can be written for the volume of the conductor as follows:

$$\vec{F} = \int_V \vec{J} \times \vec{B} dv \quad (3)$$

where $\vec{J}$ is the current density vector, and V is the volume of conductor. Using the average value of the gap flux density in radial direction as B_{gap} , it can be rewritten as

$$F = SV_c B_{gap} J \quad (4)$$

where V_c is the bulk coil volume including non-conducting parts, and $0 < S < 1$ is the space efficiency. If the VCA is operated in the moving magnet mode, mass of the iron cores add up to the moving mass as

$$m_{tot} = m_{init} + n \times m_{core} \quad (5)$$

where m_{init} is the initial mass of the moving body, n is the number of VCAs attached, m_{core} is the mass of the iron core, and m_{tot} is the final mass. The equivalent mass per actuator can be calculated as

$$m_{eq} = m_{tot}/n \\ = m_{init}/n + m_{core}. \quad (6)$$

As a sample case, $m_{init} = 0.725$ kg and $n = 2$ are selected. Different values can be used for different applications. Using (4) and (6), acceleration per current density can be expressed as

$$\sigma_{acc} = \frac{SV_c B_{gap}}{m_{eq}}. \quad (7)$$

Maximum current density for continuous operation can be set using the 700 circ·mils/A rule [28], which is equivalent to $J_{max} = 2.819$ A/mm². Hence, maximizing σ_{acc} maximizes the acceleration for all selections of coil wire gauge, as $\max(F/m_{eq}) = \max(\sigma_{acc}) \times J_{max}$. When there is a steady-state bearing force, like in a flexure bearing [15], σ_{acc} can be modified as

$$\sigma_{acc}^* = \frac{SV_c B_{gap}}{m_{eq}} - \frac{F_b}{J_{max} m_{eq}} \quad (8)$$

where F_b is the maximum bearing force along the stroke length. Points with $\sigma_{acc}^* < 0$ should be excluded from design as the maximum bearing forces cannot be overcome in those even when the maximum current is applied.

2) *Motor Constant*: Heat dissipation of the coil is given as

$$Q = I^2 R. \quad (9)$$

Total resistance of the wire can be expressed as

$$R = \frac{\rho N l_t}{A} \quad (10)$$

where ρ is the resistivity, N is the number of coil turns, l_t is the average loop length, and A is the cross-sectional area of the conductor in the wire. The average loop length l_t can also be expressed as

$$l_t = \frac{SV_c}{NA}. \quad (11)$$

Using (11) in (10), expression for wire resistance can be rewritten as

$$R = \frac{\rho SV_c}{A^2}. \quad (12)$$

The standard definition of motor constant for rotary electric servomotors [2], can be reformulated to relate instantaneous force produced to the instantaneous heat generation as

$$K_M = \frac{F}{\sqrt{Q}}. \quad (13)$$

Motor constant in $(N/\sqrt{W})$ allows current terms to be cancelled out, and gives an indication of how much force is produced per unit square root of heat dissipation rate. For the optimal case, it has to be maximized. Substituting (12) and the relationship, $I = J \cdot A$, in (9), and using the result in (13) together with (4), motor constant can be expressed as

$$K_M = \sqrt{\frac{S V_c B_{\text{gap}}^2}{\rho}}. \quad (14)$$

When there is a steady-state bearing force, (14) can be used without modifications, as it uses force instead of acceleration. It has been shown in [19] that a similar figure of merit defined as $K_{\text{fm}} = F/(\sqrt{Q} \cdot m_{\text{core}})$ for MMA would be independent of actuator size as long as proportions between dimensions are conserved. It can be related to K_M as $K_{\text{fm}} = K_M/\sqrt{m_{\text{core}}}$.

B. Determination of δ and h_3

δ should be set to a minimum, as it reduces the available volume for the coil V_c , hence decreasing the optimization objectives. $\delta = 0.5$ mm is set in this paper. As δ is related to thermal expansion of components, as much as the manufacturing/assembly errors, it should be proportionally bigger for designs with higher r_3 .

The lumped reluctance term R_{st3} used in MEC analysis suggests that h_3 should be increased in order to increase the flux in the circuit. However, it has been determined from FEA that increasing h_3 reduces B_{gap} by diverting flux to the leakage paths. Increasing the h_3 also increases the mass of the iron core. However, if it is too small, saturation occurs in the flux path, and hence it is set to $h_3 = 6$ mm with a margin for saturation.

C. Performance Topologies With r_1 and p

After the variables δ and h_3 are determined, the optimization procedure reduces to the search for an optimal pair $\{r_1, p\}$ such that the linear combination of parameters σ_{acc} and K_M , as given in (7) and (14), respectively, is maximized. Instead of simply combining the two objectives with a weighting factor, the designer's engineering judgment also needs to be incorporated into the process. Therefore, a performance-topology-based approach has been followed, as detailed below. For this, optimization parameters have to be evaluated for each candidate set.

Variables that show up in the optimization objectives are set as follows. ρ is set to the resistivity of copper, V_c can be calculated from the varying coil dimensions, while m_{eq} can be calculated from iron core dimensions and material densities. The cross section of the coil wire with layers of insulation and bonding material is shown in Fig. 4. The figure represents the most conservative estimation, as the layers could be slightly shifted with respect to each other to improve packing.

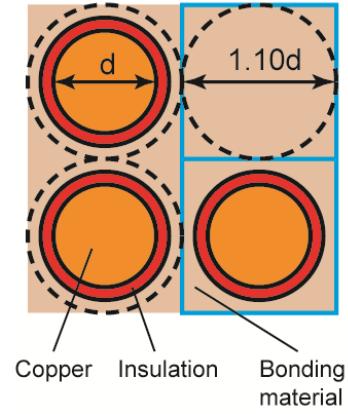


Fig. 4. Coil diagram related to space efficiency.

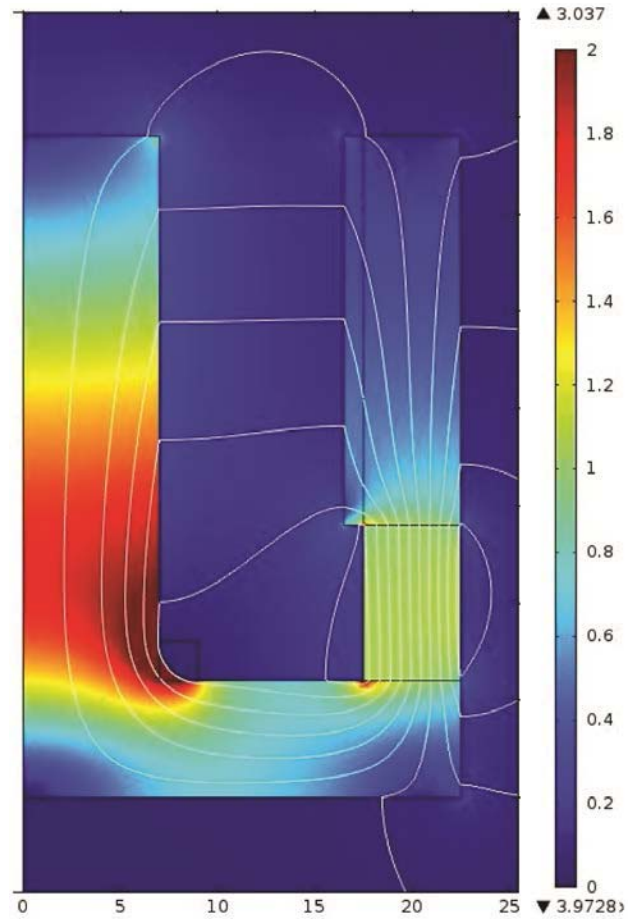


Fig. 5. COMSOL FE analysis for the variable pair ($r_1 = 7$ mm, $p = 1$ mm).

The average of standard minimum and maximum increase in the diameter of bare copper wire due to single (Type 1) and heavy (Type 2) bonding for wires AWG#18 to AWG#30 is found to be 10.8% (rounded to 10% in Fig. 4) [27]. The space efficiency corresponding to the 10% average increase in bare wire diameter can be calculated as

$$S = \frac{N A l_t}{N [d(1 + 0.10)]^2 l_t} = \frac{\pi d^2 / 4}{[d(1 + 0.10)]^2} \approx 0.65 \quad (15)$$

where d is the diameter of the conducting wire. To obtain the value of B_{gap} , COMSOL FE model is updated

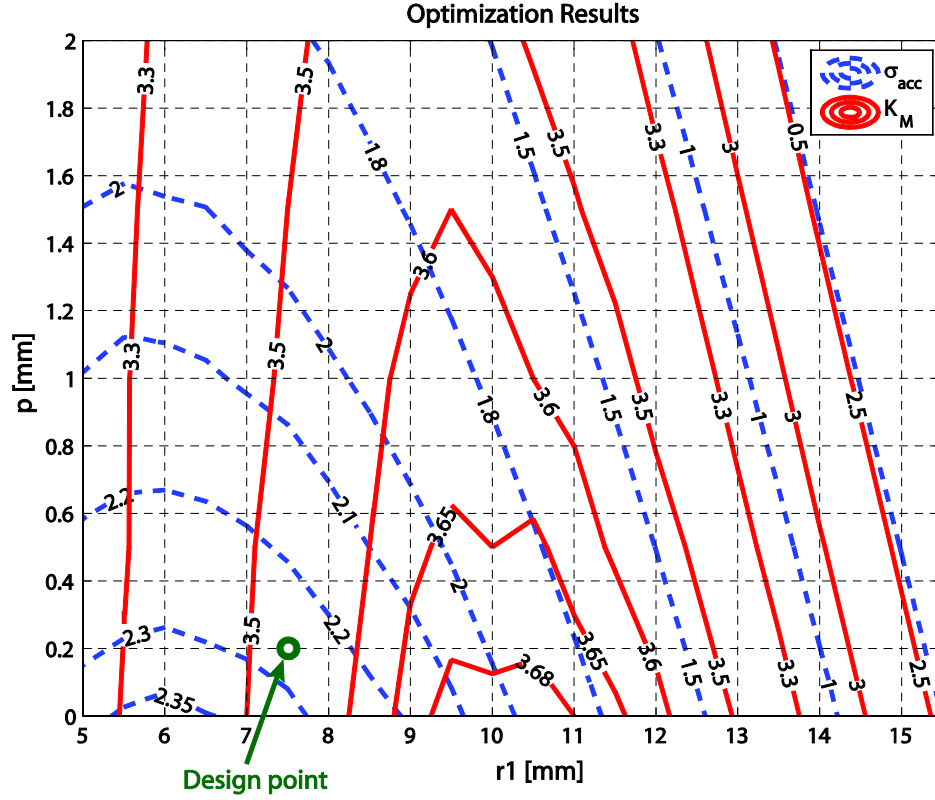


Fig. 6. Performance topologies.

for each variable set and the magnetic flux densities are simulated. A screenshot from the FE analysis of the VCA for $\{r_1 = 7 \text{ mm}, p = 1 \text{ mm}\}$ is shown in Fig. 5.

Around 100 simulations have been conducted to map performance parameters to discrete values of r_1 and p . The resulting performance topologies are charted in terms of the optimization parameters as shown in Fig. 6.

From the examination of the topology, the chosen design point is $\{r_1 = 7.5 \text{ mm}, p = 0.2 \text{ mm}\}$. $r_1 = 7.5 \text{ mm}$ is chosen, as it corresponds to a point with desirable balance between highest σ_{acc} and K_M points, horizontally. Performance topologies suggest the feature given by p to be discarded. However, this feature is kept using the small $p = 0.2 \text{ mm}$, in case its effect on the reduction of leakage fluxes is underestimated in FEA. In practice, tolerance on p can be expected as $\pm 0.050 \text{ mm}$ for all p , as it assumes the tolerance of the turning operation that produces the pole piece with $r_2 - p$ inner radius. The design point $\{r_1 = 7.5 \text{ mm}, p = 0.2 \text{ mm}\}$ has $\sigma_{acc} = 2.27 \text{ m}^3/\text{As}^2$ and $K_M = 3.55 \text{ N}/\sqrt{W}$. The slight advantage given to σ_{acc} is favorable, because motion bandwidth is generally more critical than heat dissipation. In the FEA results of the VCA with the final parameters, regions with $B > 2.0 \text{ T}$ were minimal. If these regions were bigger, as might be observed for smaller r_1 , FEA has to be repeated using the full profile of the $B-H$ curve, which takes non-linearity and saturation into account. If not, actual VCA magnetic fields would be seen to be proportionally lower than the FEA predictions.

TABLE III
ELECTRICAL PARAMETERS FOR DIFFERENT WIRE GAUGES

AWG# (d) (mm)	N (turns)	$L_{50\%}$ (mH)	R (Ω)	I_{max} (A)	e_{DC} (V)	e_{250} (V)
18 (1.024)	139	0.7	0.22	2.32	0.5	2.6
20 (0.813)	220	1.7	0.56	1.46	0.8	4.1
22 (0.643)	353	4.4	1.42	0.91	1.3	6.5
24 (0.511)	559	11.1	3.57	0.58	2.1	10.3
27 (0.361)	1120	44.5	14.3	0.29	4.1	20.6
30 (0.254)	2258	181	58.3	0.14	8.3	41.5

IV. COIL DESIGN AND ELECTRICAL CHARACTERISTICS

A. Number of Coil Wire Turns, N

The number of coil wire turns can be expressed as

$$N = \frac{4S(y_2 - y_1)h_1}{\pi d^2} \quad (16)$$

where d is the diameter of the conductor in the wire and S is the space efficiency. The range of S is calculated from (15) as 61%–69% for single (Type 1) and 56%–64% for heavy (Type 2) bonding wires between AWG#18–30 [27]. In manufacturing, the values can turn out to be slightly lower or higher, depending on the quality of packing. Number of coil turns N calculated for different wire gauges with $S = 0.65$ are presented in Table III.

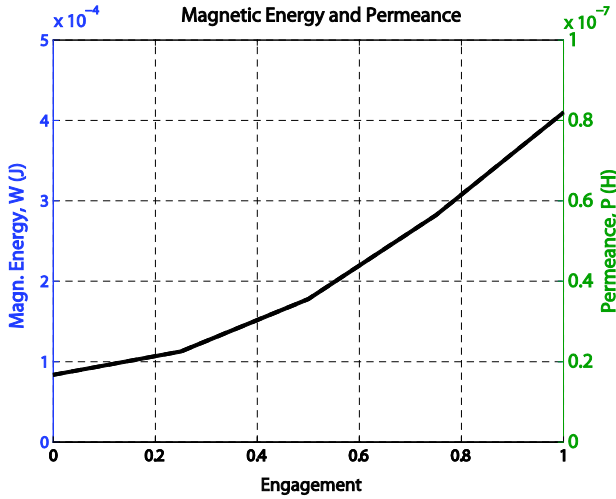


Fig. 7. Magnetic energy and permeance versus coil engagement. $N = 100$ turns.

B. Coil Inductance, Resistance and Time Constant

The coil inductance is given as [21]

$$L = N^2 P \quad (17)$$

where P is the overall permeance of the magnetic circuit around the coil. The magnetic energy stored in the inductor is given as [21]

$$W = \frac{1}{2} L I^2. \quad (18)$$

Using (17) in (18), the permeance of the magnetic circuit P can be found from magnetic energy as

$$P = \frac{2W}{N^2 I^2}. \quad (19)$$

The magnetic energy is simulated using FE software using an example coil with $N = 100$ turns, considered to be running 1-A current. This is just a test case for the identification of the overall permeance of the magnetic circuit, which is independent of coil and current selection. Simulated magnetic energy and corresponding overall permeance values for varying levels of coil engagement to the core are shown in Fig. 7.

Inductance values calculated using the overall permeance P at a nominal engagement of 50% are presented in Table III. Resistance values calculated according to (10) are also presented in Table III. The time constant τ can be expressed as

$$\tau = \frac{L}{R} = \frac{SV_c P}{\rho l_t^2}. \quad (20)$$

Equation (20) shows that the time constant is independent of the wire gauge.

C. Power Supply

The equation for instantaneous voltage supply required for actuation is given as

$$e = IR + \frac{dI}{dt}L + I \frac{dL}{dz} \frac{dz}{dt}. \quad (21)$$

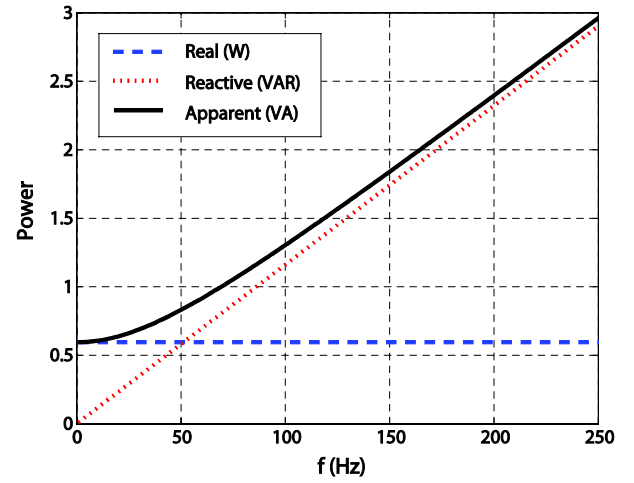


Fig. 8. Power components in the AC case as a function of current frequency.

The maximum current values, as presented in Table III, are found by

$$I_{\max} = A J_{\max}. \quad (22)$$

1) *DC Case*: For the DC case, the maximum voltage requirement and power consumption can be written as

$$e_{DC} = I_{\max} R = \frac{J_{\max} \rho S V_c}{A} \quad (23)$$

$$P_{DC} = I_{\max}^2 R = J_{\max}^2 \rho S V_c. \quad (24)$$

Corresponding values of e_{DC} for different wire gauges is presented in Table III. P_{DC} is found to be 1.19 W for all wire gauges.

2) *AC Case*: When driving the VCA with a sinusoidal current, $I(\omega) = I_{\max} \cos(\omega t)$, amplitude of the voltage supply required is

$$e_{AC}(\omega) = I_{\max} \sqrt{R^2 + L^2 \omega^2}. \quad (25)$$

e_{AC} at 250 Hz is presented in Table III as e_{250} . The complex power during ac operation can be expressed as [29]

$$\begin{aligned} P_{AC} &= \frac{1}{2} I_{\max}^2 R + j \omega \frac{1}{2} I_{\max}^2 L \\ &= \frac{1}{2} J_{\max}^2 \rho S V_c + j \frac{1}{2} \omega J_{\max}^2 \frac{S^2 V_c^2}{l_t^2} P. \end{aligned} \quad (26)$$

It can be observed from (26) that both real and reactive power components are independent of the wire gauge. Power components are plotted in Fig. 8 against ω .

3) *Effect of Back-EMF*: We can approximately define a voltage constant, $K_v = dL/dz$, by approximating the change in L using the two points of coil engagement, 25% and 75%, according to (17) as

$$K_v = -N^2 \frac{P_{0.75} - P_{0.25}}{h_1}. \quad (27)$$

K_v values calculated for the sample set of wires are given in Table IV.

The maximum acceleration achievable is given as

$$a_{\max} = \sigma_{\text{acc}} J_{\max}. \quad (28)$$

TABLE IV
EFFECT OF BACK-EMF ON THE VOLTAGE DEMAND

AWG# (d) (mm)	$ K_v $ (V/(m/s))	e_v (V)
18 (1.024)	0.033	0.027
20 (0.813)	0.082	0.043
22 (0.643)	0.211	0.070
24 (0.511)	0.530	0.110
27 (0.361)	2.126	0.220
30 (0.254)	8.645	0.444

The velocity limit (assuming a triangular velocity profile) is

$$v_{\max} = \sqrt{h_1 a_{\max}} = 0.36 \text{ m/s.} \quad (29)$$

Additional voltage requirements, $I_{\max}|K_v|v_{\max}$, for different wires are presented in Table IV. The maximum power demand due to back-electromotive force (EMF) is given by

$$\begin{aligned} P_{b-\text{emf}} &= I_{\max}^2 |K_v| v_{\max} \\ &= J_{\max}^2 v_{\max} \frac{S^2 V_c^2}{l_f^2} \left(\frac{P_{0.75} - P_{0.25}}{h_1} \right). \end{aligned} \quad (30)$$

$P_{b-\text{emf}}$ is found to be 0.064 W for all wire gauges.

D. Final Design of the Coil

From Tables III and IV, the dominant voltage term is observed to be e_{250} . All voltage terms (e_{250} , e_{DC} , e_v) can be reduced by selecting thicker wire gauge. However, $I_{\max}$ also increases with thicker wire and can exceed available current. AWG#24 wire was used in the experimental VCAs. For the experiments, the power op-amp Texas Instruments OPA549T used has enough voltage/current available to actuate both the selected and the other wire gauges in the sample set.

V. EXPERIMENTAL STUDY OF THE VCA DESIGN

In order to verify the FE analysis and formulations, two samples of the finalized VCA design were built in-house. The permanent magnet and steel parts were gradually brought to contact by removing a set of thin wood plates one by one from the interface. The permanent magnet and the steel core were aligned using a nylon sleeve manufactured to a snug fit.

A. Verification of the Magnetic Analysis Method

Using a Lake Shore 410 hand-held Gaussmeter, the gap flux density of the iron core was measured as shown in Fig. 9.

COMSOL FE prediction and measured values for different points along the radius of the iron core are compared in Table V. While evaluating predictions of the FEA, half of the probe thickness, $t/2 = 0.75$ mm, is offset from the evaluation points on inner and outer walls.

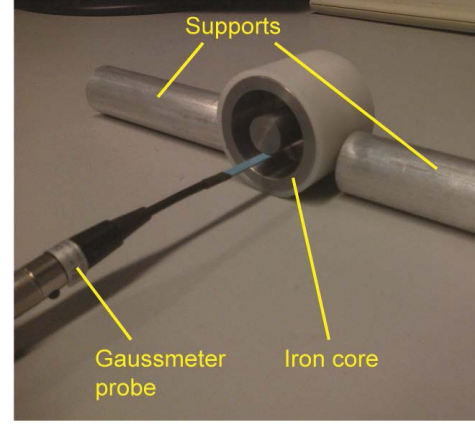


Fig. 9. Experimental measurement of flux densities.

TABLE V
COMPARISON OF FE AND EXPERIMENTAL FLUX DENSITIES

Loc.	FEA (T)	VCA 1		VCA 2	
		Exp. (T)	FEA Err. (%)	Exp. (T)	FEA Err. (%)
Inner wall	0.230	0.212	+8.5	0.213	+8.0
Outer wall	0.120	0.107	+12.1	0.107	+12.1
Middle	0.156	0.140	+11.4	0.142	+9.9

During FE simulation, the magnetic property of iron (AISI 1018 steel) was approximated by the constant relative permeability, $\mu_r = 4000$. The NdFeB magnet with Grade 42 has been modeled by a linear $B-H$ curve of slope, $\mu_r = 1.1$, and remanent magnetic flux density of $B_r = 1.28$ T. Deviations of the measurements from FEA results are on average 10%, which might have originated from the non-linear magnetic properties of iron, mismatch of magnet properties, miscalculation of leakage fluxes, manufacturing or measurement errors. For the practical application of the VCA, this discrepancy is acceptable.

B. Verification of Force and Performance Formulations

Coils were built using AWG#24 ($d = 0.511$ mm) NEMA standard MW-29C magnet wire with PB#1 (Polyvinyl Butyral) bond coat activated by alcohol (solvent bonding). A 0.5-mm aluminum layer is used in the coil inner diameter to improve strength. Hence, effective coil inner and outer radii have become $y_1 = 8.5$ mm and $y_2 = 16.8$ mm, respectively.

The VCA design was implemented into an ultraprecision motion stage, used as the experimental setup for force measurements as shown in Fig. 10.

At one end of the shaft, the iron core is mounted. The coil, which engages to the iron core, is fixed on the mounting bracket and is fully engaged into the core. The other end of the shaft contacts a conical cap that is screwed onto the FUTEK LCM200 load cell. When the coil is powered, flat end of the shaft presses the load cell against the spacer, which is supported by the bracket. For force measurements,

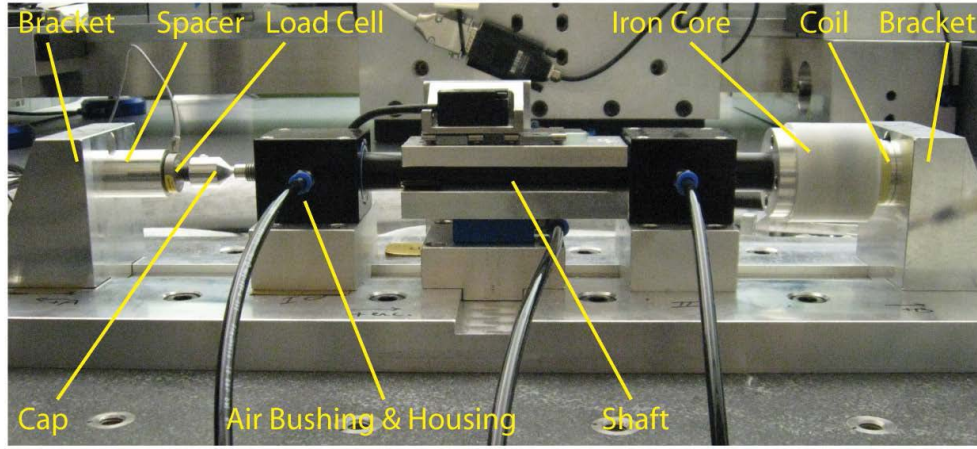


Fig. 10. Experimental setup for force measurement.

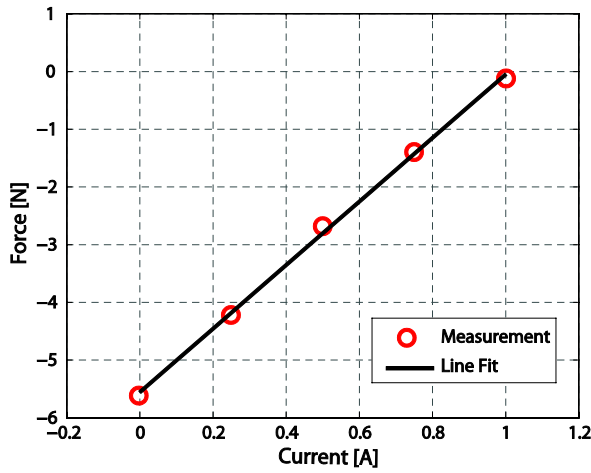


Fig. 11. Force response of the VCA.

the coil is supplied with steady current via the Texas Instruments OPA549T power amplifier and the response of the load cell is recorded. The measurement results are shown in Fig. 11.

The least-squares line fit in Fig. 11 indicates a measured force factor $K_f = 5.508 \text{ N/A}$. Using the definition of current density in the force expression of (4), the analytical expression for force factor is given as

$$K_f = \frac{S V_c B_{\text{gap}}}{\pi d^2/4}. \quad (31)$$

Using the average of experimentally obtained gap flux densities at the middle of the gap from Table V, $B_{\text{gap}} = 0.141 \text{ T}$, the analytical force factor is found as $K_f = 5.907 \text{ N/A}$, which is 7.2% greater than the measured one. This may have resulted from the actual space efficiency S , being lower than the predicted one, and from the possible measurement errors in registering the gap flux density. Nevertheless, there is good agreement between the predicted and actual force factors.

VI. CONCLUSION

In this paper, detailed design and optimization of a VCA for precision motion applications has been presented.

The achievable bandwidth of the prospective motion system is maximized by the maximization of the stage's acceleration capability. Also, heat dissipation, which becomes a serious source of inaccuracy by causing thermal deformations, is minimized. The optimization was carried out by considering only the bulk shape of the coil, thus making wire gauge selection only a concern from the standpoint of electrical drive system limitations. Effects of different wire gauge selections on the current and voltage supply demands from the electrical drive were studied. The design procedure presented can be easily adapted to the design of VCAs with different stroke length, force capacity, overall shape, and configuration from the one studied in this paper.

APPENDIX

EFFECT OF REDUCED OUTER DIAMETER OF POLE PIECES AND BASE PLATE CHAMFER

In VCAs designed for loudspeakers, often the pole pieces are slightly smaller in diameter compared with the permanent magnet [23]. This feature reduces end leakage, and hence can increase gap flux density. In order to account for this design feature, an extra parameter c can be introduced, as shown in Fig. 12. A chamfer on the base plate toward the outer diameter is also introduced in Fig. 12, for reducing leakage.

FEA is carried out for the finalized VCA ($r_1 = 7.5 \text{ mm}$, $p = 0.2 \text{ mm}$) for various values of c , with and without chamfer. Results are presented in Table VI.

For the no chamfer case, the maximum increase in B_{gap} happens for $c = 0.75 \text{ mm}$, which is 1.3% higher than the original with $c = 0 \text{ mm}$. As c is increased further, gains in B_{gap} start diminishing due to the reduced effective diameter of the magnet, and eventually fall below the original value. When base plate chamfer is employed, biggest improvement in B_{gap} is observed for $c = 0.50 \text{ mm}$, 4.0% higher than the original design.

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